



Cambridge IGCSE™

MATHEMATICS

0580/21

Paper 2 Extended

October/November 2022

MARK SCHEME

Maximum Mark: 70

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2022 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **7** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Maths-Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Abbreviations

cao – correct answer only

dep – dependent

FT – follow through after error

isw – ignore subsequent working

oe – or equivalent

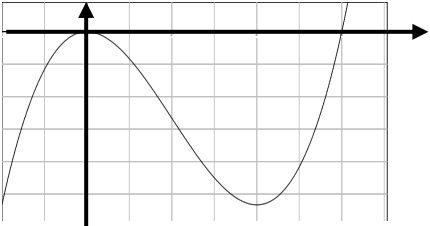
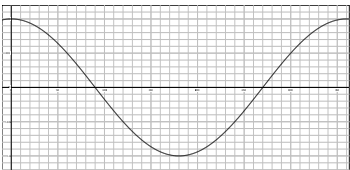
SC – Special Case

nfw – not from wrong working

soi – seen or implied

Question	Answer	Marks	Partial Marks
1	Any multiple of 72	1	
2	7 h 10 min	1	
3	$\frac{4}{25}$ cao	2	M1 for $\frac{32}{200}$ oe
4	140, 60	2	M1 for $\frac{200}{(7+3)} \times k$ where $k = 1, 7$ or 3
5	54	2	M1 for $180 - 71 - 55$ oe or B1 for 55 or 125 in a relevant correct position on the diagram
6	442	2	M1 for $\frac{100-15}{100} \times 520$ oe or B1 for 78
7(a)	a, b, c, d	1	
7(b)	6	1	
8(a)	243	1	
8(b)	$4n + 9$ oe final answer	2	B1 for $4n + k$ or $jn + 9, j \neq 0$ or for correct answer seen then spoilt
9	$\frac{2}{6} + \frac{5}{6}$ oe	M1	i.e. correct fractions with common denominator $6k$
	$1\frac{1}{6}$ cao	A1	
10	$2x^9$ final answer	2	B1 for kx^9 or $2x^k$ as final answer or $2x^9$ spoiled
11	$[x =] 4$ $[y =] -1$	2	B1 for each
12(a)	6	1	
12(b)	8	2	M1 for $\left(\frac{2}{3}\right)^2$ or $\left(\frac{3}{2}\right)^2$ oe seen
13(a)	2.8 oe	1	
13(b)	175	2	M1 for a correct relevant area calculation e.g. $(15 - 5) \times 14$ or $\frac{1}{2} \times 5 \times 14$ oe or better

Question	Answer	Marks	Partial Marks
14	Rotation (5, 3) 90° clockwise oe	3	B1 for each
15	71.6 or 71.61 to 71.62	3	M2 for $\frac{\text{angle}}{360} = \frac{26-8-8}{2\pi \times 8}$ or better or M1 for $\frac{\text{angle}}{360} \times 2\pi \times 8$ oe
16	[u =] 20 [v =] 52 [w =] 108 [x =] 36	4	B1 for each
17	$5x^{625}$ final answer	2	B1 for final answer kx^{625} or $5x^k$ or correct answer spoiled
18	12.7 or 12.68 to 12.69	4	M3 for $\frac{7\sin 115}{\sin(180-115-35)}$ or B2 for 8.03... seen OR B1 for [angle C =] 30 M2 for $\frac{7\sin 115}{\sin(\text{their angle } C)}$ or M1 for $\frac{\sin 115}{BC} = \frac{\sin(\text{their angle } C)}{7}$ oe
19	$2x^3 - 5x^2 - 4x + 12$ final answer	3	B2 for correct expansion of the three brackets unsimplified or for simplified four-term expression of correct form with three terms correct or B1 for correct expansion of two of the three given brackets with at least three terms out of four correct
20(a)	$(1+x)(1-y)$ final answer	2	B1 for $1+x-y(1+x)$ or $1-y+x(1-y)$
20(b)	$2x(x+3y)(x-3y)$ final answer	3	B2 for $2x(x^2-9y^2)$ or correctly factorising into two brackets e.g. $(2x^2+6xy)(x-3y), (x^2-3xy)(2x+6y)$ or B1 for $2(x^3-9xy^2)$ or $x(2x^2-18y^2)$ or for $(x+3y)(x-3y)$

Question	Answer	Marks	Partial Marks
21	Correct sketch with maximum at origin and minimum in fourth quadrant 	2	B1 for any cubic with exactly 2 distinct turning points
22(a)	Correct sketch  Correct sketch to go through (0, 1), (360, 1) and (180, -1)	2	To go through (0, 1) and close to (360, 1) and reasonably close to (180, -1) B1 for correct cosine curve shape through (0, 1)
22(b)	120, 240	2	B1 for each or for two values with sum of 360
23	$\frac{144}{w}$ oe	3	M2 for $y = \frac{k}{w}$ oe or M1 for $x = cw^2$ or for $y = \frac{j}{\sqrt{x}}$ oe
24	4 nfw	2	M1 for $39 + 0.5$ or $36 - 0.5$ or better seen $39 - 0.5$ or $36 + 0.5$
25(a)(i)	$\frac{3}{4}$ oe	1	
25(a)(ii)	45	1	FT $60 \times$ <i>their</i> (a)(i) correctly evaluated
25(b)	$\frac{47}{66}$ oe	4	M3 for $1 - \left(\frac{5}{12} \times \frac{4}{11} + \frac{4}{12} \times \frac{3}{11} + \frac{3}{12} \times \frac{2}{11} \right)$ oe or M2 for $\left(\frac{5}{12} \times \frac{4}{11} + \frac{4}{12} \times \frac{3}{11} + \frac{3}{12} \times \frac{2}{11} \right)$ oe or $\left(\frac{5}{12} \times \frac{4}{11} + \frac{5}{12} \times \frac{3}{11} + \frac{4}{12} \times \frac{3}{11} \right)$ oe or M1 for $\frac{5}{12} \times \frac{4}{11}$ or $\frac{5}{12} \times \frac{3}{11}$ or $\frac{4}{12} \times \frac{3}{11}$ or $\frac{3}{12} \times \frac{2}{11}$ oe If 0 scored, SC1 for $\frac{47}{72}$ oe

Question	Answer	Marks	Partial Marks
25(c)	5	2	M1 for correct trial to at least two balls one of which is not green